An Opportunistic Routing Strategy for Circulation Flow-Guided Nano-Networks

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ABSTRACT

The advancement in nanotechnology is expected to accelerate the development of intra-body nano-networks. With the aim of tackling the energy deficiency and positioning problems of intra-body nano-networks, this paper takes the unidirectional characteristic of blood circulation system into consideration and proposes a circulation flow-guided opportunistic routing protocol (COR). The COR protocol with two-layer network structure enhances data transmission performance by the designed candidate relay selection (CRS) forwarding algorithm, which also improves the utilization and life cycles of the nano-nodes. This protocol is simulated under the 3D circulation flow vessel model combined with flow-guided movement model in Nano-Sim platform, and proved its bio-security as well as its superiority.

CCS CONCEPTS

• **Computer systems organization → Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

KEYWORDS

Intra-body Nano-Networks, Opportunistic Routing, Flow-Guided, Candidate Relay Selection

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1 INTRODUCTION

Intra-body nano-networks are envisioned as the most promising technology to increase the potential application of the detection and treatment of disease. Different kinds of disease, neuronal disorder or cancers etc. [6, 19], are believed to have strong relation with blood indices. Therefore, intra-body nano-networks enable new medical developments such as health monitoring, vital sign collection, and data transmission within the human body [15, 23]. To satisfy the requirements of customized medical detection or treatments, it is crucial to investigate feasible network topology for nano-devices working in blood circulation system. The number of required nano-devices is an important factor under the consideration, while lessening the influence of nano-devices to the human body and controlling the expenditure for whole working system [3]. Besides, the movements of nano-nodes are mostly affected by blood flow, and only if nano-nodes within the communication range of the gateway can transmit the data transmission successfully. Therefore, packet forwarding will be adopted to achieve high communication efficiency. Moreover, it is also necessary to reduce energy consumption of nano-nodes as much as possible, while improving the effective transmission rate.

Under the consideration of the above-mentioned challenges of intra-body flow-guided nano-networks, the contributions of this paper are summarized as follows: (i) We propose a candidate relay selection (CRS) forwarding algorithm to determine forwarding priority to improve the performance of forwarding, (ii) a circulation flow-guided opportunistic routing protocol (COR) is proposed based on CRS forwarding algorithm, which reduces the energy consumption while increasing the transmission rate in circulation flow guidance environment, (iii) the above-mentioned protocol is investigated in Nano-Sim simulation platform under different conditions and parameter values.

The COR protocol defines a two-layer network structure consisting of two distinct nano-devices, the top layer group and the base layer group. The base layer group consists of nano-nodes randomly moving under overall orientation guided by blood flow, and its main function is to collect biological information in the blood vessel. The top layer group consists of gateways responsible for receiving data from nano-nodes and sending information to external macro-devices, wearable devices or smartphones etc. Since the gateway needs to communicate with the external macro-devices, it is generally fixed in the vein close to superficial skin. The base layer group can transmit and receive data packets through THz band inside the human body. This two-layer network structure benefits in two aspects. Firstly, it can reduce the bio-affinity consideration and also help reduce the difficulty of network implementation. Secondly some complex computing tasks can be done directly in the

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nano-gateway rather than extra data packets forwarding through nano-nodes.

The rest of this paper is organized as follows: Section 2 presents relevant research in nano-networks. In Section 3, we propose the COR protocol and introduce the design of the COR protocol in detail. In Section 4, we simulate the COR protocol under different number of nodes, and the bio-affinity consideration of the COR protocol. Finally, in Section 5, we summarize the research content of this paper.

2 RELATED WORK

Traditional modulation has difficulty to adapt the low-energy characteristics of intra-body nano-networks [5]. Due to the limitation of extremely small size of nano-devices, it is recommended to use the Terahertz (THz) frequency band to communicate between nano-devices. Furthermore, it is difficult for nano-scale antennas through THz band communication to achieve desired theoretically large bandwidth [4, 10]. Some researches on modulation and demodulation methods proposed new modulation to substitute traditional modulation of continuously sending carrier waves. Time Spread On-Off Keying (TS-OOK) in [11] is a pulse-based signal modulation widely used in nano-networks for its great adaptation.

Nano-nodes in circulation flow-guided nano-networks will capture energy from flowing blood to compensate the defect of limited energy storage and ensure durative operation of these nano-nodes. Because of the insufficient energy, nano-node might turn to sleep mode until energy is sufficient to restart its work. Therefore, the analysis of energy capture ability of piezoelectric generators in nano-nodes and the energy consumption model of nano-nodes sending data packets is extremely required. To prolong the working life of nano-nodes, some research studied the energy harvesting technology of nano-nodes. Jornet et al. [17] proposed a method to convert mechanical energy into electrical energy using piezoelectric zinc oxide nano-wire-arrays. According to the values used in [2, 5], for the 500 um^2 nanometer generator, the expression of the energy capture and estimation of the energy storage in nano-nodes are concluded. The high path loss of intra-body nano-networks put forward high requirements for energy harvesting ability of nano-nodes, and this issue is preliminarily researched in [16, 22].

In addition, when nano-nodes communicate with each other in the flow-guided liquid environment, the surroundings will slightly heat up, which may cause negative influence on blood cells. For the safety and bio-affinity consideration, we calculate the temperature effect of networks by modeling and evaluate the influence based on subsequent simulation results. In intra-body nano-networks, the electronic waves radiated by nano-nodes during data transmission will cause the molecules in liquid environment to be excited and vibrate, causing part of the energy of the propagating waves converted into kinetic energy [1]. This causes the rise in temperature of biological cells, which is influenced by the number of nano-nodes transmitting data and the time cells are exposed to electronic waves. To ensure the safety of cells or other organs in the human body, it is necessary to accurately calculate the temperature rise range during communication. Cumulative Equivalent Minutes at 43°C (CEM_{43}) is an accepted unit of measure for thermal exposure (temperature and exposure time), which is closely related to thermal damage in various tissues[18].

Research in [12] introduced the basic physical mechanism by which molecules absorb noise. Reference [7] proposed an approximate closed-form expression for the distribution of the temperature increase of the red blood cells, using the Laplace transform and line-of-sight probability. The molecular absorption of the channel model and the health issues corresponding to the communication in the THz band is evaluated in article [8]. Moreover, a mathematical framework to compute absorption from different types of molecules, as well as scattering of both the cells and the medium for intra-body communication in the THz band is presented in article [20].

3 DESIGN OF COR PROTOCOL

In dynamic nano-networks, motion laws and routing path of mobile nodes are unpredictable. Opportunistic routing protocol is a very appropriate solution for dynamic nano-networks, because it selects the forwarding direction of the route in real-time according to the actual situation [21]. In circulation flow-guided nano-networks, after verifying surrounding nano-nodes through broadcast, the nano-node determines the forwarding priority of the neighbor nano-nodes according to a lightweight calculation before forwarding. This will be repeated until the data is forwarded to the nano-gateway, and finally producing an optimal path in the process of random movement of locations. Opportunistic routing protocol can reduce the energy consumption compared to flood routing protocol, and can also greatly improve the utilization efficiency of circulation flow-guided nano-networks.

3.1 3D Circulation Flow-Guided Vessel Model

A 3D circulation flow-guided vessel model for closed-form modeling of the circulation flow-guided environment in human body is proposed, and which consists of two steps. Firstly, we model the blood vessel as a circulatory pipeline, and restrict nano-nodes to operate and move within this pipeline. The second is to set movement rules for nano-nodes that move irregularly with the blood in the blood vessel.

In a real environment, by sending and receiving electromagnetic signals, nano-nodes determine whether they are within the communication range of each other. However, under the conditions of simulation, we use a three-dimensional coordinate system to verify real-time position and communication range. In this three-dimensional coordinate system of the blood vessel, the x axis represents the advancing position of nano-nodes in the circulation blood vessel, the y axis represents the left and right offset position, and the z axis represents the up and down offset position. The judgment of neighbor nodes between node a and node b are conducted by calculating the Euclidean distance between each other as

$$d(a,b) = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2 + (z_a - z_b)^2} \tag{1}$$

where (x_a, y_a, z_a) is the three-dimensional coordinate of node a, (x_b, y_b, z_b) is the coordinate of node b, *d(a,b)* is the distance between node a and node b. If *d(a,b)* is less than the signal transmission distance of the nano-nodes, node a and node b are neighbor nodes to each other, vice versa. In the 3D circulation flow-guided vessel model, when the node moves along the x-axis to the end of the blood vessel, its position will turn to the starting point and start a new cycle. Since we use a section of pipeline to simulate the circulation, considering whether the two nodes are currently in the same turn of blood flow, calculation of the distance between two nano-nodes should be modified from Euclidean distance formula to the following formula

$$D = \begin{cases} \min\left(d\,(a,b), \sqrt{(x_a + x_{\max} - x_b)^2 + (y_a - y_b)^2 + (z_a - z_b)^2}\right), x_b \geqslant x_a \\ \min\left(d\,(a,b), \sqrt{(x_b + x_{\max} - x_a)^2 + (y_b - y_a)^2 + (z_b - z_a)^2}\right), x_b < x_a \end{cases} \tag{2}$$

where (x_a, y_a, z_a) is the coordinate of node a, (x_b, y_b, z_b) is the coordinate of node b, D is the distance between node a and node b in the 3D circulation flow-guided vessel model, X_{max} represents the maximum length of the x-axis.

In the flow-guided environment of the human body, the movement of nano-nodes in the blood vessel is a process of forward movement and random direction changing. This process can use the Gauss-Markov movement model [13] to simulate, which was first proposed as a simulation applied to personal communication systems. Since the movement of nano-nodes is limited within the blood vessel, when nano-node reaches the vessel wall in the left-right direction of the y-axis or the up-down direction of the z-axis, its radial moving direction will change to angle of reflection. Meanwhile, its forward direction will maintain unchanged. In 3D circulation flow vessel model, there are no limitations for nano-nodes moving forward along the x-axis. Only when reaching the end of pipeline in simulation, its position will turn to the starting point and start a new cycle. Thus, the position of nano-node a can be given as

$$a = \begin{cases} (x_a, y_a, z_a)\,, x_a \leqslant\ X_{\max} \\ (x_a - X_{\max}, y_a, z_a)\,, x_a > X_{\max} \end{cases} \tag{3}$$

3.2 Time Relative Position Model

Nevertheless, the deployment of traditional position sensors (such as Global Position System) is impossible due to the size of nano-nodes, which makes mobile nano-nodes blind to position and mobility information. When selecting a forwarding nano-node, it is impossible to determine the geographic location of the nano-node and gateway. The proposed time relative position model helps to determine the data packet forwarding direction towards the gateway and form the further forwarding path.

In time relative position model, each nano-node carries an *Index* value to mark the relative time when it moves in the blood vessels. The *Index* value increases by 1 in each energy capture time slot. Larger *Index* value means longer run-time of nano-nodes, smaller *Index* for shorter run-time. Distribution of the *Index* values of nano-nodes in the blood vessel model reflects the flow direction of blood, as shown in Fig. 1. The nano-node passing through the communication range of the nano-gateway will receive a probe packet with a Time To Live (TTL) value of 1, which is broadcasted by the nano-gateway in a time period of T_t. After receiving the probe packet, the nano-node will reset its *Index* value to 0, so as to start the next cycle of *Index* value counting. The TTL of the probe packets is set to 1 to prevent network congestion caused by too many probe packets in the working system. Time relative position model

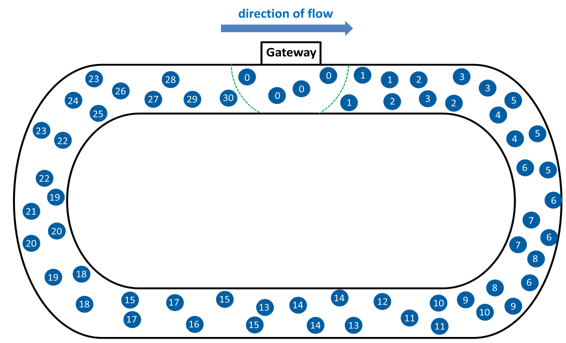

Figure 1: Time Relative Position Model of Nodes in COR

enables the representative of the relative position of nano-nodes in blood vessels, which provides a clear direction for the data packet forwarding, as shown in Fig. 2.

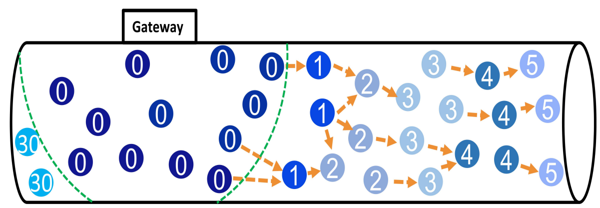

Figure 2: Data Forwarding Direction in COR

3.3 Candidate Relay Selection

An important step in circulation flow-guided nano-networks is to select the candidate relay node of the sending node. According to the above-mentioned time relative position model, the distribution of the *Index* values of nano-nodes in the blood vessel model reflects the flow direction of blood. Higher *Index* value means a longer run-time and ahead in position. Therefore, nano-node will select neighbor nodes with higher *Index* values than itself to join its CRS when sending data packets, as shown in Fig. 3

When the sending node selects CRS, it first broadcasts a probe data packet carrying its own *Index* value, and nano-nodes that can receive the probe data packet are neighbor nodes of the sending node. Neighbor nodes judge whether their remaining energy can guarantee the energy consumption in subsequent forwarding operations as follows:

$$E_{re} > E_{r-p} + E_{s-t} + E_{s-p} + E_{r-t} + E_{r-ack} \tag{4}$$

where E_{r-p} represents the energy consumed by receiving data packets, E_{s-t} represents the energy consumed by sending probe packets, E_{s-p} represents the energy consumed by sending data packets, E_{r-t} represents the energy consumed by receiving gateway probe packets or neighbor nodes responding to probe packets, and E_{r-ack} represents the energy consumed by receiving neighbor node ACK response. If the energy of the neighbor node meets the energy condition and its own *Index* value is greater than the *Index* value in the probe data packet, the neighbor node sends a response

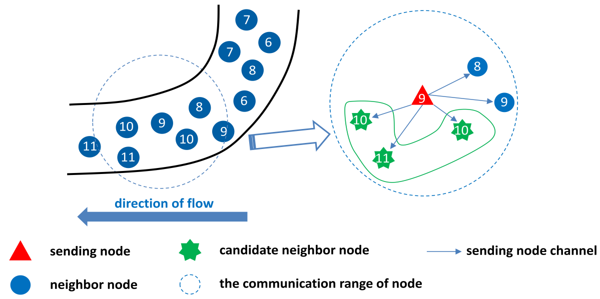

Figure 3: CRS Selection of Sending Nodes

data packet to the sending node. The response data packet contains information such as the ID value, *Index* value, and remaining energy of the neighbor node. If the energy of the neighbor node does not meet the energy condition or its own *Index* value is less than or equals to the *Index* value in the probe packet, the neighbor node maintains silence.

The sending node incorporates all surrounding neighbor nodes that have sent response data packets into its own CRS. If the CRS is empty, the sending node stops forwarding and continues to move with the blood to wait for the next CRS selection. The CRS selection period of the nano-node is T_{CRS}. If there is only one candidate node in the CRS, the candidate node will be chosen as the next-hop data packet forwarding node, and the sending node will forward the data packet to this candidate node. If there are multiple candidate nodes in the CRS, the sending node needs to calculate the forwarding priority according to the CRS priority algorithm, then forward its data packets to the candidate node with the highest priority in the CRS. After the data packet is successfully forwarded, the sending node discards this data packet, and the selected node becomes the new sending node. This process cycles in turn, until nano-gateway is found in the communication range of sending node. Then the data packet is forwarded to the nano-gateway, and the data packet forwarding process is finished.

3.4 CRS Priority Algorithm

When there are multiple nano-nodes in the CRS, it is necessary to calculate their respective forwarding priorities for efficient forwarding. The priority value V_{Pri_i} of nano-node i is calculated as

$$V_{Pri_i} = \lambda_{index} \left| Index_i - Index_s \right| + \lambda_{re} E_{re-i} + \gamma N_i, \tag{5}$$

where λ_{index}($0<\lambda_{index}<1$), λ_{re}($0<\lambda_{re}<1$) and γ are system preset parameters. $Index_i$ is the *Index* value of the candidate node *i*, $Index_s$ is the *Index* value of the sending node, E_{re-i} is the remaining energy of candidate node *i*, and N_i is ID of the candidate node. Which λ_{index} and λ_{re} satisfies the condition of

$$\lambda_{index} + \lambda_{re} = 1. \tag{6}$$

The main reference indicators of the priority value V_{Pri_i} are the *Index* value and the remaining energy E_{re} of the candidate node. The ID of the candidate node is added to prevent a conflict in which candidate nodes have same calculation results in priority determination. After the priority values V_{Pri_i} of all candidate nodes are calculated, the sending node will sort the priority values V_{Pri_i} of each candidate node from large to small. Each candidate node i will get a priority sequence number Pri_i according to the sorting result. The smaller Pri_i value is, the higher forwarding priority of the candidate node is. If the number of candidate nodes is *n*, the value range will be 0 to *n*-1. After obtaining the priority sequence numbers Pri_i of candidate nodes, the sending node forms a correspondence table among the IDs of all candidate nodes and their Pri_i. The sending node broadcasts the corresponding table together with the carried data packet.

3.5 CRS Back-off Forwarding Algorithm

Neighbor nodes within the communication range of sending node will receive the above-mentioned corresponding table and the data packet. The neighbor node will first look up the priority sequence number corresponding to its own ID in the corresponding table. If its own ID is not in the corresponding table, it will not operate. But if it is in the table, it will get its own back-off slot δ according to its own priority sequence number. Each candidate node will receive the data packet only when its own back-off slot ends. The back-off time slot $\delta(i)$ for the candidate node *i* is

$$\delta(i) = \begin{cases} 0, Pri_i = 0 \\ 2(t_t + t_s)Pri_i + t_t, Pri_i > 0 \end{cases} \tag{7}$$

where t_s is the time required for the nano-nodes to send or receive the entire data packet, and t_t is the time for transmission of the data packet in the channel. The calculation of t_s and t_t are

$$t_s = \left(8\left(S_p + S_t\right) + S_h\right)\left(T_s + T_p\right) - T_s \tag{8}$$

$$t_t = \frac{R_{node}}{c} \tag{9}$$

where S_p is the size of the original data packet (in Byte), S_t is the size of the priority sequence number corresponding table (in Byte), S_h is the size of fixed header of the transmitted data (in Bit), T_s is the pulse gap, T_p is the pulse duration, R_{node} is the communication range of nano-nodes, and *c* is the propagation speed of electronic waves, which is the same as the speed of light.

The back-off time slot δ of the candidate node with the highest priority is 0, and this candidate node will be the pioneer to receive the data packet. As the data packet is received successfully, the candidate node will return a ACK response packet to the sending node. The sending node will broadcast this ACK response packet within the communication range, and the TTL of the ACK response packet is 1. After receiving the ACK response packet, all candidate nodes still waiting in the back-off slot will stop waiting and discard the data packets received before.

If a high-priority candidate node malfunctions or fails to receive data packets due to other reasons, it will not send back ACK response packet. Thus if a candidate node with lower priority does not receive an ACK response packet at the end of its back-off time slot, it will receive the data packet and send an ACK response packet to the sending node to complete the forwarding process. Because it is considered that possibly the candidate node with a high priority failed to receive the data packet. Broadcasting the ACK response packet by the sending node is aiming to ensure that every candidate node is in the transmission range of ACK response packet.

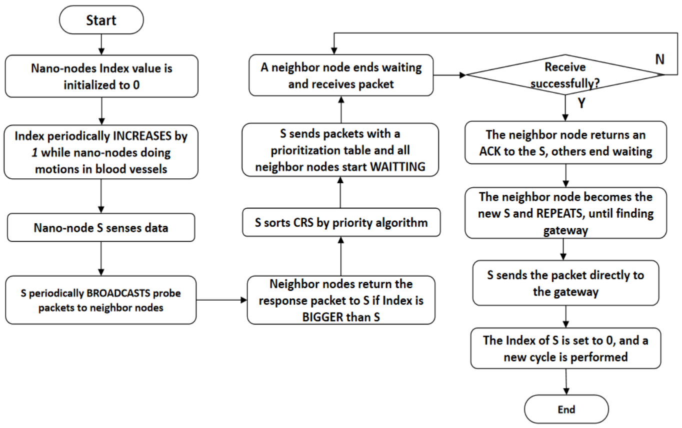

Figure 4: Flow Chart of COR Protocol

The basic working flow of the circulation flow-guided opportunistic routing protocol is described and the chart of the COR strategy is shown in Fig. 4.

4 SIMULATION ANALYSIS

In this section, in order to evaluate the performance and bio-security of circulation flow-guided nano-networks, Nano-Sim is used to simulate the nano-networks composed of randomly deployed nano-nodes [14]. In this simulation, the parameters of the pipeline in 3D circulation flow vessel model are 50cm in length and 1cm in width and height. The nano-gateway is deployed in the blood vessel at the starting position of the model. Detailed simulation parameter settings are shown in Table II. The proposed routing protocol with simulation models is compared with the Flooding routing protocol and the Random routing protocol. The specific experimental results are as follows. In 15s simulation time, nano-nodes run at least two laps, according to the length of blood vessel and the blood velocity setting in the simulation.

4.1 Successful Packet Transmission Rate

The successful packet transmission rate is one of the most important meitrics in routing protocols. In this simulation, the calculation of the successful packet transmission rate R_s is

$$R_s = \frac{N_{rec}}{\sum_{i=0}^{n} N_{sent_i}} \tag{10}$$

where N_{sent_i} is the number of data packets generated by nano-node i, and N_{rec} is the number of data packets received by the nano-gateway. The experimental results of the successful packet transmission rate are shown in Fig. 5.

As shown in Fig. 5, the proposed COR protocol has better performance on successful packet transmission rate, compared with the Flooding routing protocol and the Random routing protocol. The overall successful packet transmission rate of the three routing protocols is relatively high when there are less nano-nodes, but the successful packet transmission rate gradually decreases with the increase of the number of nano-nodes. When the choice of data packet forwarding path becomes more complex, nano-nodes

Table 1: Simulation Experiment Parameters

Parameter	Value
System parameters [9]	
Simulation time	15 s
Density of nano-nodes	$[0.5-2]$ $nodes/cm^3$
Number of nano-gateways	1
Transmission Loss Parameters	0.4
Artery size	$50cm^3$
Maximum storage energy	0.8 pJ
The time slot of Energy-harvesting	0.1 s
Energy-harvesting rate	$[0.7-0.8]$ pJ/s
The speed of blood	10 cm/s
PHY details	
Pulse energy E_p	100 aJ
Pulse duration T_p	100 fs
Pulse interval T_s	10 ps
Network Layer	
Initial TTL value	15
Message Processing Unit	
The duration of packet generation	$[2.0-5.0]$ s
The duration of probe packet generation	0.1 s
The duration of energy capture	0.1 s
The election duration of candidate nodes	0.1 s

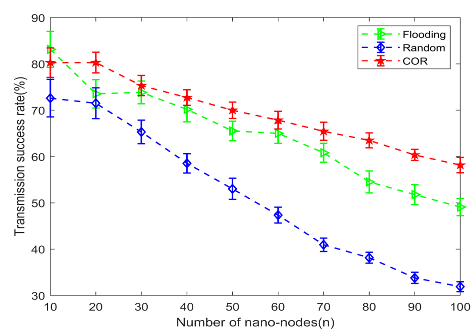

Figure 5: Successful Packet Transmission Rate

consume more energy on forwarding path choice. For the Random routing protocol, the TTL of the data packet using Random routing protocol is 15. The TTL of the data packets will consume more energy when the number of nano-nodes increased, eventually reduces the successful packet transmission rate. While using the Flooding routing protocol, nodes will send packets to all their neighbor nano-nodes, which causes more energy comsuption compared with COR protocol. Thus, it is easier for nodes to run out of energy and make transmission fail.

4.2 Average Energy Consumption

The average energy consumption of one successful packet transmission reflects the available life cycle of the circulation flow-guided nano-networks. The smaller average energy consumption refers to the longer available life cycle. The average energy consumption

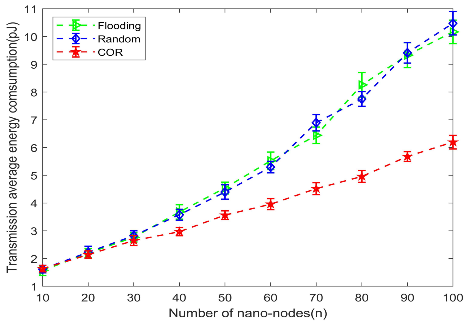

Figure 6: Average Energy Consumption

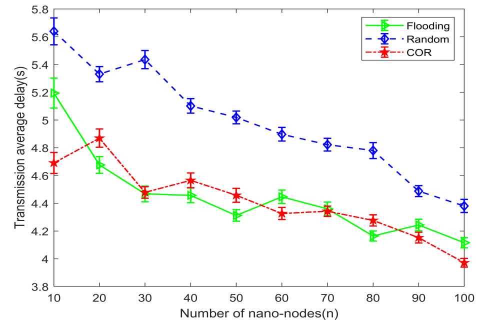

Figure 7: Average Delay of Data Packet Transmission

E_{av} can be calculated as

$$E_{av} = \frac{\sum_{i=0}^{n} E_{t_i}}{N_{rec}} \tag{11}$$

where E_{t_i} is the total energy consumed by nano-node i during the operation period, and N_{rec} is the total number of data packets received by the nano-gateway. The experimental results of the average energy consumption are shown in Fig. 6.

As shown in Fig. 6, the proposed COR protocol also has the best performance on average energy consumption. The average energy consumption of the three routing protocols increases with the increase of the number of nano-nodes, and the consumption between the Flooding routing protocol and the Random routing protocol is roughly equal. This is because the COR protocol reduces energy consumption by selective forwarding path and less packet forwarding terms.

4.3 Average Packet Transmission Delay

The average packet transmission delay represents the time interval in average from when a data packet is generated to when it is received by the nano-gateway. The smaller average packet transmission delay, the more data packets can be transmitted per unit time. In this simulation, the calculation of the average packet transmission delay TD_{av} is

$$TD_{av} = \frac{\sum_{i=0}^{n} \left(t_{rec_i} - t_{sent_i}\right)}{N_{rec_i}} \tag{12}$$

where t_{rec_i} represents the timing of the data packet i to be received by the nano-gateway, t_{sent_i} represents the timing of the data packet i to be generated, and N_{rec} represents the total number of data packets received by the nano-gateway. The experimental results of the average packet transmission delay are shown in Fig. 7.

As shown in Fig. 7, COR performs as good as Flooding routing protocol in the average packet transmission delay, better than Random routing protocol. The average packet transmission delay of the three routing protocols decreases with the increase of the number of nano-nodes. This is because nano-nodes using COR forwards the data packet to the neighbor node whose position is ahead of itself through forwarding priority influenced by *Index*, thereby shortening the transmission path and reducing the average packet transmission delay. With the increase of the number of nano-nodes, more data forwarding reduces the packet transmission delay, and the accuracy rate of forwarding choice using random protocol is also improved. Thus the performance of all three routing protocols is better.

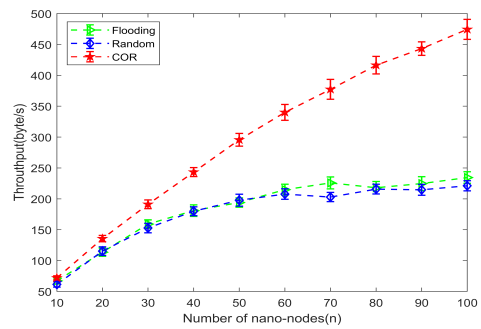

Figure 8: Throughput of Network System

4.4 Effective Throughput

The effective throughput represents the bits received by the entire nano-networks in unit time. Greater effective throughput means stronger processing capability of nano-nodes and more stable nano-networks. The effective throughput is calculated as

$$NT = \frac{\sum_{i=0}^{n} B_{rec_i}}{T_{sim}} \tag{13}$$

where B_{rec_i} represents the bits received by nano-node i in time T_{sim}, and T_{sim} represents the simulation time of the experiment. The experimental results of the effective throughput are shown in Fig. 8.

As shown in Fig. 8, COR protocol has the highest effective throughput among these three routing protocols and the Flooding routing protocol is the lowest. This is because the COR protocol has a low average packet transmission delay, thus more data packets are successfully transmitted per unit time. However, in the Flooding routing protocol, some of the nano-nodes carry the same data packets. So the number of non-repetitive data packets that are successfully transmitted per unit time is small, resulting in low node utilization and throughput.

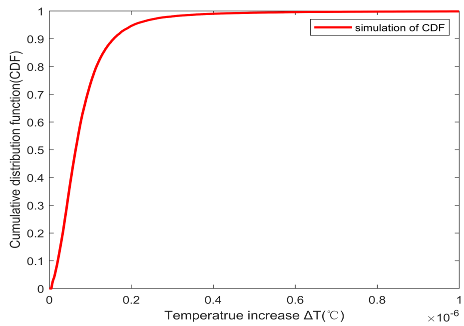

Figure 9: The CDF of Nano-Networks System

4.5 Effect of Heat Energy

Aiming to ensure the bio-security of the algorithm, we set the number of nano-nodes to 100, and do simulation 100 times to test the maximum temperature increase in the blood vessel. The experimental results are shown as cumulative distribution function (CDF) in Fig. 9. Experimental results show the range and the possibility of temperature increase.

Fig. 9 shows that up to 99% probability, the temperature increase in the blood vessel is lower than $4 * 10^{-7}$°C. This temperature will not be harmful to cells in the human body and there is no need to worry about bio-security.

5 CONCLUSION

In this paper, we proposed a circulation flow-guided opportunistic routing protocol for intra-body nano-networks. We also proposed 3D circulation flow vessel model combined with flow-guided movement model of nano-nodes to simulate the human body environment of blood vessels. The energy capture and consumption model of nano-nodes in the circulation flow-guided nano-networks is analyzed. Considering the energy capture and consumption, we proposed CRS forwarding algorithm, which also takes more conditons in intra-body research field into consideration. CRS forwarding algorithm is the combination of CRS forwarding priority calculation and CRS back-off forwarding. Finally, we simulate circulation flow-guided opportunistic routing protocol in Nano-Sim platform and the experimental results are showed and analyzed in Sec.4. The circulation flow-guided opportunistic routing protocol performs great in successful packet transmission rate and average energy consumption etc. targets. The COR protocol fits for circulation flow-guided nano-networks. Although, in all of the simulations we did, our proposed protocol have superority compared with traditional ways, further reseaches are required for improvements from two perspective. First, implantation and removal of nano-devices is a huge challenge. Secondly, when it comes to real human body, communication between nano-nodes may be influenced by other cells in blood vessels, which also causes performence degradation of the algorithm.

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